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# Secunda: Controllable Autoregressive Generation for Adaptive Videogame Music

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Daniele Bracoloni

## Abstract

Adaptive videogame music aims to provide coherent transitions between gameplay states, shifting instantly from a relaxed exploration theme to intense combat encounters. Can a token-based generative model be conditioned by continuous acoustic control trajectories? Secunda is a token-based autoregressive transformer designed to generate 24 kHz audio conditioned on combat and tension. Operating on an EnCodec representation, the 48.5M-parameter model dynamically modulates discrete audio token prediction through Feature-wise Linear Modulation.

## 1. Introduction

Repetitive musical transitions can break player immersion in modern videogames. While traditional dynamic soundtracks rely on pre-composed stems, this project explores a generative approach using deep learning. Secunda is a token-based autoregressive transformer that models adaptive videogame music generation as a discrete sequence prediction task.

To operate at a high acoustic fidelity without exceeding computational constraints, the system processes audio using the EnCodec neural audio codec at 24.0 kbps and 75 frames per second. While the codec extracts 32 residual codebooks, Secunda predicts only the first 8. This architectural trade-off reduces sequence complexity while avoiding the severe "AUX unplugged" distortion artifacts caused by relying solely on the first 4 structural codebooks. Secunda is steered by two independent, frame-aligned continuous signals: Tension, which captures the acoustic RMS energy, and Combat Score. The Combat Score is an interpretable acoustic-activity proxy based on percussiveness, articulation, brightness, and harmonic roughness. Us-

ing Feature-wise Linear Modulation (FiLM), the 48.5M-parameter model dynamically alters its latent token predictions to match the desired acoustic trajectory. The baseline model was trained for 60 epochs on an initial Slakh pool of 800 tracks (704 training and 96 validation). It was then fine-tuned using an expanded Slakh pool of 1,395 tracks, while a separate 151-track Slakh split was retained for held-out evaluation. A final adaptation stage used videogame soundtrack material.

## 2. Related Work

Neural audio codecs provide a compact discrete representation on which generative sequence models can operate. SoundStream (Zeghidour et al., 2022) and EnCodec (Défossez et al., 2022) represent waveforms as parallel streams of discrete tokens, making autoregressive modeling computationally tractable. AudioLM (Borsos et al., 2023) demonstrated coherent audio continuations using these tokens, while MusicGen (Copet et al., 2023) introduced delayed codebook interleaving to handle residual vector quantization streams efficiently. Secunda adopts this perspective but focuses on conditioning short game-music cues with continuous acoustic controls. To inject these controls, I use Feature-wise Linear Modulation (FiLM) (Perez et al., 2018), which applies a condition-dependent affine transformation to intermediate activations, modulating the hidden representation for each codebook output head based on tension and combat trajectories.

## 3. Architecture and Training Methodology

Secunda is a causal Transformer decoder with 14 layers, 8 attention heads, a 448 embedding dimension, and a feedforward dimension of 2304 (48.5M parameters). Audio is pre-processed into mono 24 kHz waveforms and tokenized via EnCodec at 24.0 kbps, producing 32 residual codebooks at 75 Hz. To balance reconstruction quality with autoregressive complexity, the model predicts only Q0–Q7 on 10-second segments (750 frames).

Because EnCodec’s residual vector quantization is hierarchical, the 8 codebooks are serialized using a diagonal de-

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Email: Daniele Bracoloni <bracoloni.1985841@studenti.uniroma1.it>.

lay pattern: Q0 is placed at delay 0, up to Q7 delayed by 7 frames. This allows the model to predict the  $t$ -th step of codebook  $q$  conditioned on all previous timesteps and the current timestep of preceding codebooks. Missing target positions are padded with a dedicated PAD token (ID 1024), expanding the vocabulary to 1025 per head.

### 3.1. Acoustic Conditioning Proxies

Tension is computed at the codec frame rate, while the combat score is evaluated over a 5-second trailing causal window with a 1-second hop; both are aligned to 75 Hz via sample-and-hold interpolation.

**Tension** is operationalized as normalized short-time RMS energy.

**Combat Score** quantifies aggressive acoustic activity independently of pure loudness. It is a weighted sum of five normalized components: **Percussive-Energy Ratio (0.30)** via Harmonic-Percussive Source Separation; **Percussive Loudness (0.25)** capturing the RMS energy of the percussive component; **Spectral Centroid (0.20)** for acoustic brightness; **Dissonance Proxy (0.15)** measuring adjacent chroma energy; and **Note Shortness (0.10)** derived from the inter-onset interval. Both signals are normalized to  $[0, 1]$  globally to prevent feature saturation.

### 3.2. Latent Modulation (FiLM) and Training

Control trajectories are injected via FiLM (Perez et al., 2018). An input MLP projects the  $[tension, combat]$  vector into the embedding dimension. At the output stage, codebook-specific MLPs produce  $\gamma$  and  $\beta$  parameters modulating the final hidden states, allowing each residual head to react to target conditions specifically.

Training uses cross-entropy for each codebook, combined through a weighted average. To emphasize higher-index residual codebooks while prioritizing coarse structures, fixed weights of  $[4.0, 3.2, 2.7, 2.2, 2.0, 2.0, 2.0, 2.0]$  are applied to Q0–Q7. Optimization uses AdamW with a batch size of 24 and 20 random crops per track. The learning rate follows a cosine decay schedule from  $2 \times 10^{-4}$  to  $3 \times 10^{-5}$  after a 4,000-step linear warmup; subsequent fine-tuning stages used smaller learning rates with the same scheduler form.

## 4. Evaluation

All evaluations were performed on 151 held-out Slakh test tracks using 10-second segments (750 frames at 75 Hz) tokenized with EnCodec at 24.0 kbps. I compare three checkpoints: **Baseline (CP3)**, the model after initial Slakh pre-training; **Slakh FT (CP1)**, further trained on an enlarged Slakh corpus with Q2–Q7 emphasis; and **OST FT (CP2)**,

subsequently fine-tuned on Skyrim and Witcher 3 soundtracks. Table 1 summarizes the results. The Baseline exhibits the highest loss. Slakh FT improves every codebook and reduces weighted test loss from 5.54 to 5.25, showing that the model benefits from additional Slakh training data. OST FT retains most of this gain but increases weighted Slakh loss slightly to 5.32, consistent with adaptation toward the target game-music domain (Figure 1).

Table 1. Per-codebook test loss on the held-out Slakh split.

| Codebook      | Baseline | Slakh FT    | OST FT |
|---------------|----------|-------------|--------|
| Q0            | 3.29     | <b>3.01</b> | 3.14   |
| Q1            | 4.34     | <b>4.02</b> | 4.15   |
| Q2            | 4.92     | <b>4.53</b> | 4.64   |
| Q3            | 5.39     | <b>4.99</b> | 5.08   |
| Q4            | 5.77     | <b>5.39</b> | 5.46   |
| Q5            | 6.01     | <b>5.71</b> | 5.76   |
| Q6            | 6.15     | <b>5.97</b> | 6.00   |
| Q7            | 6.23     | <b>6.13</b> | 6.15   |
| Weighted avg. | 5.54     | <b>5.25</b> | 5.32   |

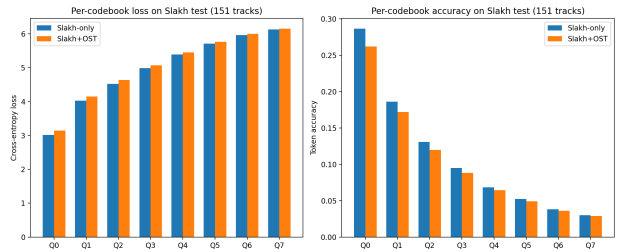


Figure 1. Per-codebook loss comparison between Slakh FT (CP1) and OST FT (CP2).

During development, I observed that models trained solely on Q0–Q3 produced severely degraded audio with spectral gaps, confirming that higher codebooks carry essential residual detail. Increasing loss weights on Q4–Q7 improved perceived fidelity, although validation losses for these codebooks remained a bottleneck. Generated audio exhibits noticeable artifacts despite the conditioning visibly responding to tension and combat trajectories. Future work will focus on expanding the dataset and integrating symbolic representations to stabilize the residual modeling.

## 5. Conclusion

In summary, Secunda demonstrates controllable autoregressive generation of short game-music cues via continuous tension and combat conditioning. Residual codebooks remain the principal fidelity bottleneck; larger datasets and longer training would be the most direct next steps.

## 6. Statement on the use of AI

I used AI tools primarily during the training and evaluation stages of this project. In particular, I relied on an AI assistant to help design training and tokenizer scripts, explore different model configurations and loss weightings, and suggest possible improvements based on validation metrics. The AI also assisted with debugging code, preparing experiment notebooks, and refining parts of the report text, including checking syntax and English usage.

The core ideas behind the project, including the overall architecture, and the design of the combat score as a custom acoustic proxy, are my own. The combat score components and weights were chosen by me through iterative listening tests and discussion with a domain expert (my brother, who works in audio production). AI tools were used only to help translate this design into Librosa-based feature extraction code and to check implementation details.

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